

Programming In C

Control Statements

Objectives of these slides:

- to introduce the main kinds of C control flow

Control Structures

- There may be situations where the programmer requires to alter normal flow of execution of program or to perform the same operation a no. of times.
- Various control statements supported by c are-
 - Decision control statements
 - Loop control statements

Decision Control Statements

- Decision control statements alter the normal sequential execution of the statements of the program depending upon the test condition to be carried out at a particular point in program.
- Decision control statements supported by c are:-
 - if statement
 - if-else statement
 - Else if Ladder
 - Nested If
 - switch statement

Decision Control Statements

■ if statement

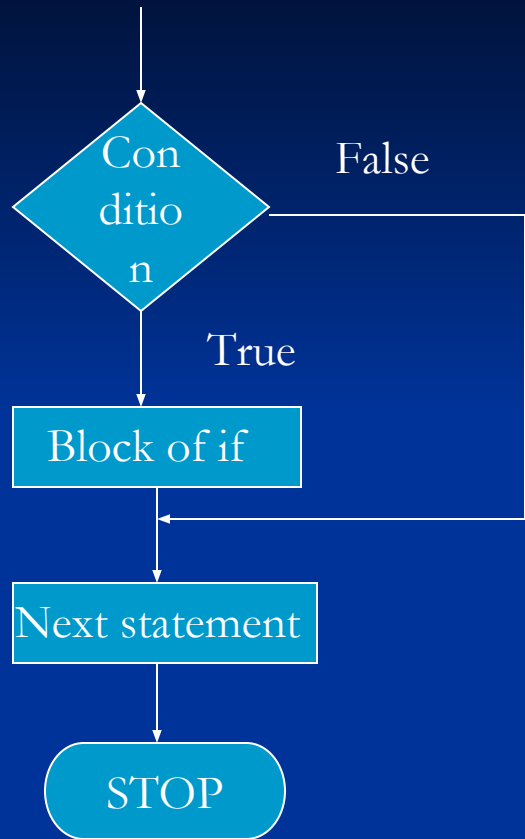
- Most simple and powerful decision control statement.
- It executes a statement or block of statements only if the condition is true.

- Syntax:

if (condition)	if (condition)
{	{
statement (s);	statement 1;
}	statement 2;
Next statement;	}
	statement 3;

- In above syntax : if condition is true only then the statements within the block are executed otherwise next statement in sequence is executed.

■ Flowchart



```
/* Program to check whether a no. is even */
```

```
# include<stdio.h>
```

```
void main()
```

```
{
```

```
int num;
```

```
printf("enter the number");
```

```
scanf("%d",&num)
```

```
if(num%2==0)
```

```
{
```

```
printf("\n Number is even");
```

```
}
```

```
printf(" End of program");
```

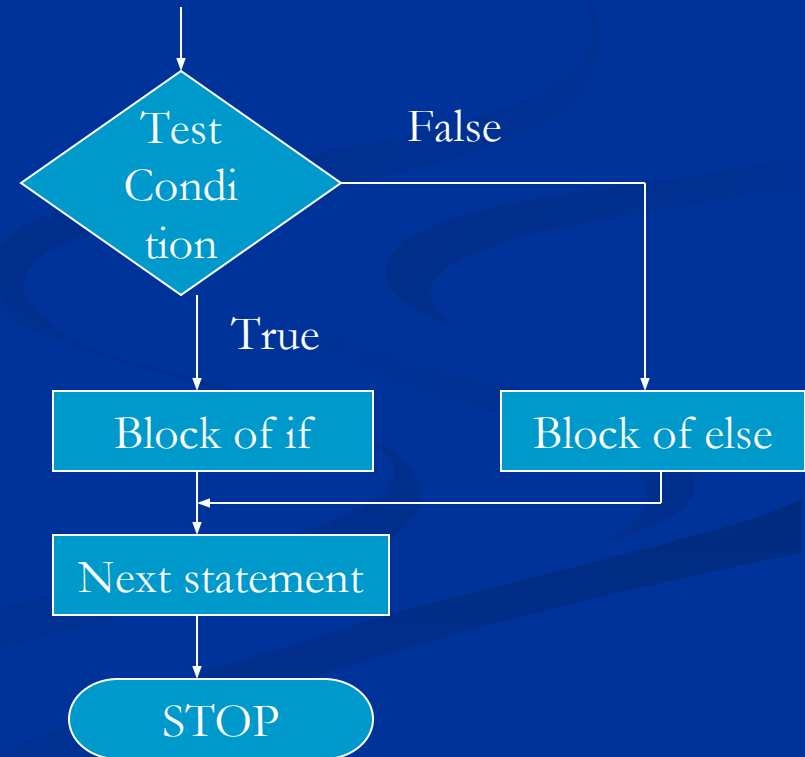
```
}
```

if – else statement

- In case of if statement, the block of statements is executed only when the condition is true otherwise the control is transferred to the next statement following if block.
- But if specific statements are to be executed in both cases (either condition is true or false) then if – else statement is used.
- In if – else statement a block of statements are executed if the condition is true but a different block of statements is executed when the condition is false.

- Syntax: if (condition)

```
{  
    statement 1;  
    statement 2;  
}  
else  
{  
    statement 3;  
}
```



Exercise: WAP to check whether a given no. is even or odd?

Nested if – else statement

- When an entire if-else is enclosed within the body of if statement or/and in the body of else statement, it is known as nested if-else statement.
- The ways of representing nested if –else are-

```
if (condition1)
{
    if (condition2)
        statement 1;
    else
        statement 2;
}
else
    statement 3;
```

```
if (condition1)
{
    if (condition2)
        statement 1;
    else
        statement 2;
}
else
{
    if (condition 3)
        statement 3;
    else
        statement 4;
}
```



```
if (condition1)
    statement 1;
else
{
    if (condition2)
        statement 2;
    else
        statement 3;
}
```

If- else- if ladder

- In a program involving multiple conditions, the nested if else statements makes the program very difficult to write and understand if nested more deeply.
- For this ,we use if-else-if ladder.

■ Syntax: if (condition1)
statement1;
else if(condition2)
statement2;
else if(condition3)
statement 3;
else
default statement;

